LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	18
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Flow Separation Stalling, behavior of aircraft before and after stall
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Flow Separation Stalling, behavior of aircraft before and after stall. Factors effecting stalling speed, high incidences stall, high speed stall, G stall, wing tip stall, Basic stalling speed.
5. Specific Objectives	6.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight simulation stall recovery procedures; Safety: Ensure simulator brake is engaged; Instructor: Monitor student input on control surfaces
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary cause of an aerodynamic stall?
(b) Define the critical angle of attack.
(c) How does increasing weight affect the basic stalling speed?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Flow Separation Mechanism • Behavior Before Stall • Behavior After Stall • Basic Stalling Speed Factors	L & Ds	PPT, IB	15 min
(b)	• High Incidence Stall Characteristics • High Speed Stall Dynamics • G Stall Effects • Wing Tip Stall Identification	L & Ds	PPT, PDF	20 min
(c)	• Recovery Techniques • Prevention Strategies • Aerofoil Section Impact	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a high incidence stall and a high speed stall.
(b) Describe the specific behavior of an aircraft immediately after a wing tip stall occurs.
(c) List three factors that influence the basic stalling speed of an aircraft.

12. Home Assignment and Follow Up	Prepare a brief note comparing the stall characteristics of a high-lift aerofoil section versus a low-lift section, referencing the concepts covered in LP 19 regarding power requirements.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Level flight performances, power available and required for jet and propeller aircrafts. Best climb speeds.	Refer to the master syllabus reference list	LP 19

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Sub-Lieutenant, P No. 12345	Senior Instructor, Lieutenant, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min